

Embryonic induction and cell division

EMBRYONIC INDUCTION AND CELL DIVISION

The stages of genomic activity that unfold in the fertilized zygote from cleavage onwards are not yet fully elucidated. An admittedly simplistic but useful approach likens developmental processes to a series of binary choices (see Fig. 8.1), the first being the choice between trophoblast and inner cell mass, which specifies if the future tissue will be placenta and fetal membranes or embryo.

Bioinformatic databases use embryological ontologies wherein the lineage of organs and tissues to be retraced to the earliest time (see **Commentary 2.1**). They provide a temporal and spatial framework against which the sequential activation of genes preceding changes in cell and tissue morphology can be matched. They also provide a means of viewing all of the complex partonomy, lineage data and temporal information occurring within embryos within a specific timeframe and of linking these to adult anatomical tissue types. As our knowledge of the complexity of early developmental events increases, ontological databases will assist the correlation of embryological information: current limitations are discussed in **Bard (2012)**.

Our knowledge of the early events that occur in a range of mammalian species and our interpretation of the significance of particular sequences of gene expression is limited. The upregulation of early genes appears to reflect the time period of implantation. In the mouse, implantation occurs one day after blastocyst formation, thus genes relevant to implantation are expressed early, whereas in cattle, implantation occurs after 14 days, by which time gastrulation has occurred and neurulation has been initiated (**Oron 2012**).

Within the developing blastocyst, the position of cells relative to each other leads to epigenetic changes. The acquisition of cell polarity is fundamental, because this specifies apical and basolateral domains, cell surface specializations, the position and nature of cell:cell junctions and the concomitant positioning of intracellular organelles and cytoskeletal elements, which in turn leads to the location of future deposition of extracellular proteins. The two cell phenotypes, epithelial and non-epithelial cells, are specified during cleavage and early compaction within the zona pellucida. All relevant genes required to maintain these cell shapes and behaviours are upregulated at these very early stages. During the formation of trophoblast, hypoblast (primitive endoderm) and epiblast, the cells respond locally to their environment, specifically to other cells that are positioned lateral and basal to them. The genes operating at these early times in the mouse embryo are given in **Takaoka and Hamada (2012)**.

Cell populations within the embryo interact to provide the developmental integration and fine control necessary to achieve tissue-specific morphogenesis. In the early embryo, such interactions may occur only if particular regions of the embryo are present, e.g. signalling centres or organizers. As the embryo matures, some interactions tend to occur between adjacent cell populations, e.g. epithelium and mesenchyme, and later between adjacent differentiating tissues, e.g. between nerves and muscle, or between muscle and skeletal elements. The interactions between adjacent epithelia and underlying connective tissue continue throughout embryonic and fetal life and extend into postnatal life. In the adult, these interactions also permit the metaplastic changes that tissues can undergo in response to local environmental conditions.

Tissue interactions result in changes or reorganization of one or both tissues, which would not have occurred in the absence of the tissue interactions. The process of tissue interaction is also called induction, i.e. one tissue is said to induce another. The ability of a tissue to respond to inductive signals is called competence, and denotes the ability of a cell population to develop in response to the environments present in the embryo at that particular stage. After a cell population has been induced to develop along a certain pathway, it will lose competence and become restricted. Once restricted, cells are set on a particular pathway of development; after a number of binary choices

(further restrictions), they are said to be determined. Determined cells are programmed to follow a process of development that will lead to differentiation. The determined state is a heritable characteristic of cells and is the final step in restriction. Once a cell has become determined, it will progress to a differentiated phenotype if the environmental factors are suitable.

The process of determination and differentiation within embryonic cell populations is reflected by the ability of these populations to produce specific proteins. Primary proteins (colloquially termed housekeeping proteins) are considered essential for cellular metabolism, whereas proteins synthesized as cells become determined, those specific to the state of determination, are termed secondary proteins; for example, liver and kidney cells, but not muscle cells, produce arginase. Fully differentiated cells produce tertiary proteins, which no other cell line can synthesize, e.g. haemoglobin in erythrocytes. The range of housekeeping, regulatory, and tissue-specific proteomes in adult cells is presented in the Human Protein Atlas (www.proteinatlas.org).

As populations of cells become progressively determined, they can be described within a hierarchy of cellular development as transiently amplifying cells, progenitor cells, stem cells and terminally differentiated cells.

Transiently amplifying cells Transiently amplifying cells undergo proliferative cell mitosis and produce cells that are equally determined. At some stage, and as a result of an inductive stimulus, these cells will enter a quantal cycle that culminates in a quantal mitosis. This will result in an increase in the restriction of their progeny, which continue to undergo proliferative mitoses at a progressive level of determination. The quantal mitosis corresponds to the time of binary choice when the commitment of the progeny is different from that of the parent.

Progenitor cells Progenitor cells are already determined along a particular pathway. They may individually follow that differentiation pathway, or may proliferate and produce large numbers of similarly determined progenitor cells that subsequently differentiate; neuroblasts or myoblasts are examples of progenitor cells.

Stem cells Either individually or as a population, stem cells can both produce determined progeny and reproduce themselves (see **Commentary 1.2**). It is generally believed that stem cells undergo asymmetric divisions, in which one daughter cell remains as a stem cell, while the other proceeds along a differentiation pathway; in marked contrast, proliferative cell division may be symmetrical, producing derived cells with an identical determination. Human embryonic stem cells (hESCs) are pluripotent cells that can be derived from the inner cell mass of human blastocysts *in vitro*; or obtained surplus to *in vitro* fertilization fertility programmes; or created from oocytes donated and fertilized for that purpose. Although not yet achieved, it is hoped that hESCs can be coaxed down particular pathways under appropriate pharmaceutical conditions to produce differentiated cells that will be effective in reversing some degenerative diseases (e.g. dopamine-producing neurones for Parkinson's disease, insulin-producing islet cells for diabetes), or to replace acutely damaged tissues (e.g. motor neurones for acute spinal cord injury, cardiomyocytes in acute myocardial infarction). Proof of principle has been demonstrated in some animal models, and multipotent haemopoietic progenitor stem cells from human umbilical cord blood are now used as an alternative treatment to bone marrow transplantation for the treatment of some inherited genetic disorders (thalassaemia) and blood malignancies (leukaemias).

Terminally differentiated cells By virtue of their extreme specialization, terminally differentiated cells can no longer divide, e.g. erythrocytes and neurones. Apoptosis is a particular variety of terminal differentiation in which the final outcome is the death of the individual cells or cell populations. It occurs in the developing limb, where cells

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die along the pre- and postaxial limits of the apical ectodermal ridge, limiting its extent, and also between the digits, permitting their separation.

Tissue interactions

There are two types of cell and tissue interaction, namely: permissive and instructive.

In a permissive interaction, a signal from an apposing tissue is necessary for the successful self-differentiation of the responding tissue. This means that a particular cell population (or the matrix molecules secreted by the cells that it contains) will maintain mitotic activity in an adjacent cell population. Since a variety of different cell populations may permit a specific cell population to undergo mitosis and cell differentiation, no specific instruction or signal that may limit the developmental options of the responding tissue is involved; this signal therefore does not influence the developmental pathway selected and there is no restriction. The responding tissue has the intrinsic capacity to develop, and only needs appropriate environmental conditions in order to express this capacity. Permissive interactions often occur later in development, when a tissue whose fate has already been determined is maintained and stabilized by another.

An instructive (directive) interaction (induction) changes the cell type of the responding tissue, so that the cell population becomes restricted. Wessells (1977) proposed four general principles in most instructive interactions: in the presence of tissue A, responding tissue B develops in a certain way; in the absence of tissue A, responding tissue B does not develop in that way; in the absence of tissue A, but in the presence of tissue C, tissue B does not develop in that way; and in the presence of tissue A, a tissue D, which would normally develop differently, is changed to develop like tissue B.

These principles are exemplified during induction of the lens vesicle by the optic cup. An example of last-mentioned principle is the experimental association of chicken flank ectoderm with mouse mammary

mesenchyme, which results in the morphogenesis of mammary gland-like structures; chickens do not normally develop mammary glands.

Tissue interactions continue into adult life and are probably responsible for maintaining the functional heterogeneity of adult tissues and organs. This is exemplified by the complex tissue heterogeneity, with sharply compartmentalized boundaries, that occurs in the oral cavity. The junctions between the mucosa of the vestibule and the lip, and between the vermillion border and the facial skin, are distinct boundaries of specific epithelial and mesenchymal differentiation, and are almost certainly maintained by continuing epithelial–mesenchymal interactions in adult life. Perturbation of these interactions throughout the body may underlie a wide variety of adult diseases, including susceptibility to cancer and proliferative disorders.

Signalling between embryonic cells and tissues

Cellular interactions may be signalled by four principal mechanisms: direct cell–cell contact; cell adhesion molecules and their receptors; extracellular matrix molecules and their receptors; and growth factors and their receptors. Many of these mechanisms interact, and it is likely that combinations of them are involved in development. Figure 11.1 illustrates some ways by which mesenchymal cells could signal to epithelial cells. An additional set of identical mechanisms could operate for epithelial–mesenchymal cell signalling. Clearly, the complexity of these mechanisms will increase in reciprocal interactions; moreover, a single molecule may have different effects on epithelial and mesenchymal cells.

Direct cell–cell contact permits the construction of gap junctions, which are important for communication and the transfer of information between cells. The transient production of gap junctions is seen as epithelial somites are formed, between neuroepithelial cells within rhombomeres, and in the tunica media of the outflow tract of the heart.

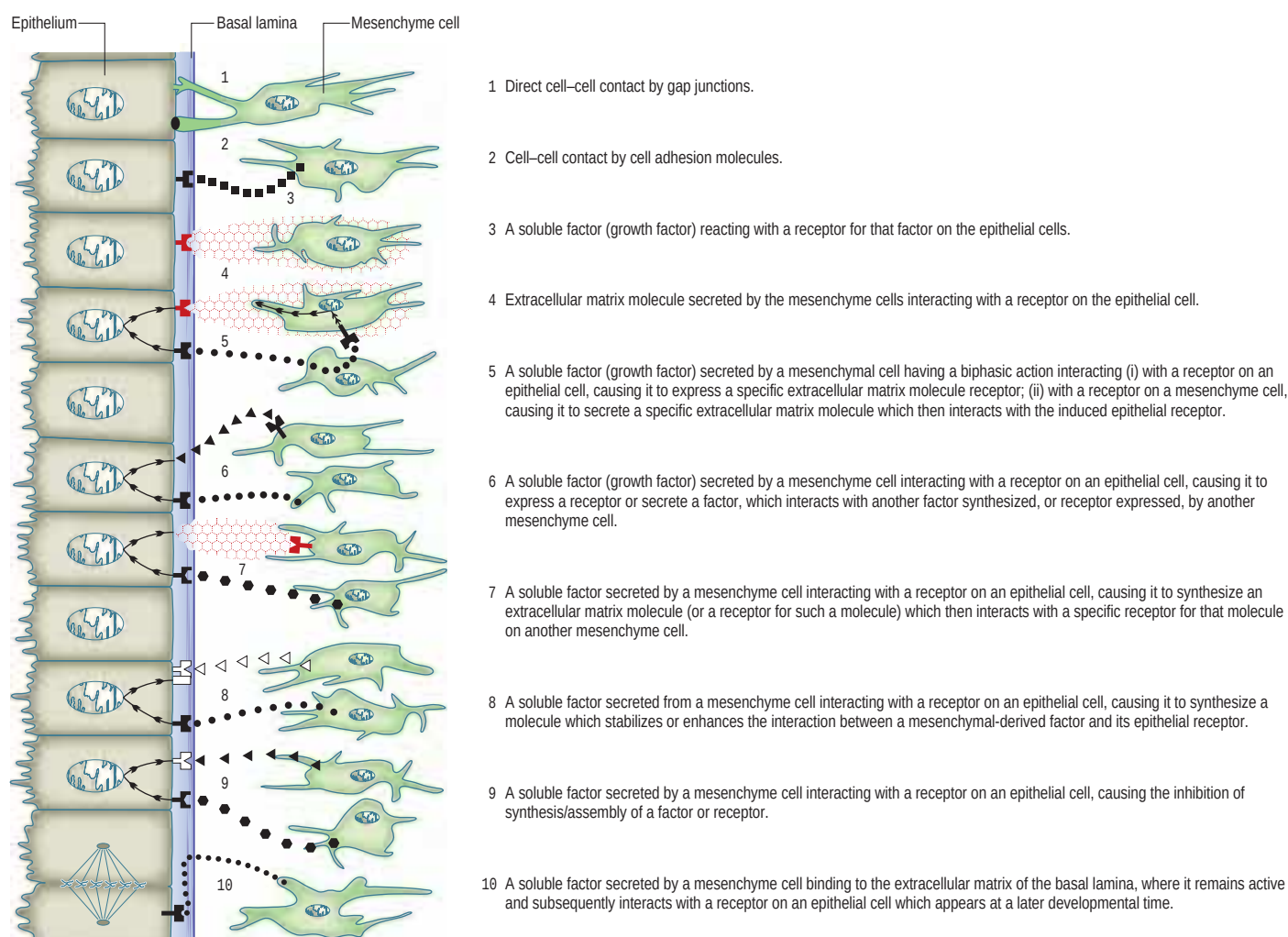


Fig. 11.1 The many ways by which mesenchyme cells could signal to epithelial cells. Precisely the same mechanisms can operate in reverse, i.e. epithelium to mesenchyme.

Endogenous electrical fields are also believed to have a role in cell–cell communication. Such fields have been demonstrated in a range of amphibian embryos, and in vertebrate embryos during primitive streak ingression. Neuroepithelial cells are electrically coupled, regardless of their position relative to inter-rhombomeric boundaries.

The spatial and temporal distribution of a variety of cell adhesion molecules has been localized in the early embryo. The appearance of these molecules correlates with a variety of morphogenetic events that involve cell aggregation or disaggregation; e.g. an early response of groups of cells to embryonic inductive influences is the expression of cadherins, calcium-dependent adhesion molecules typically found in epithelial populations. Other molecules found in the extracellular matrix, e.g. fibronectin and laminin, *inter alia*, can modulate cell adhesion by their degree of glycosylation. Self-assembly or cross-linking by matrix molecules may affect cell adhesiveness by increasing the availability of binding sites or by obscuring them.

Extracellular matrix molecules include localized molecules of the basal lamina, e.g. laminin, fibronectin, and much larger complex associations of collagen, glycosaminoglycans, proteoglycans and glycoproteins between the mesenchyme cells. Mutations of the genes that code for extracellular matrix molecules give rise to a number of congenital disorders, e.g. mutations in type I collagen produce osteogenesis imperfecta; mutations in type II collagen produce disorders of cartilage; and mutations in fibrillin are associated with Marfan's syndrome.

Growth factors are distinguished from extracellular matrix molecules. They can be delivered to, and act on, cells in a variety of ways, namely: endocrine, autocrine, paracrine, intracrine, juxtacrine and matricrine (Fig. 11.2). Many growth factors are secreted in a latent form, e.g. associated with a propeptide (latency-associated peptide) in the case of transforming growth factor beta, or attached to a binding protein, in the case of insulin-like growth factors.

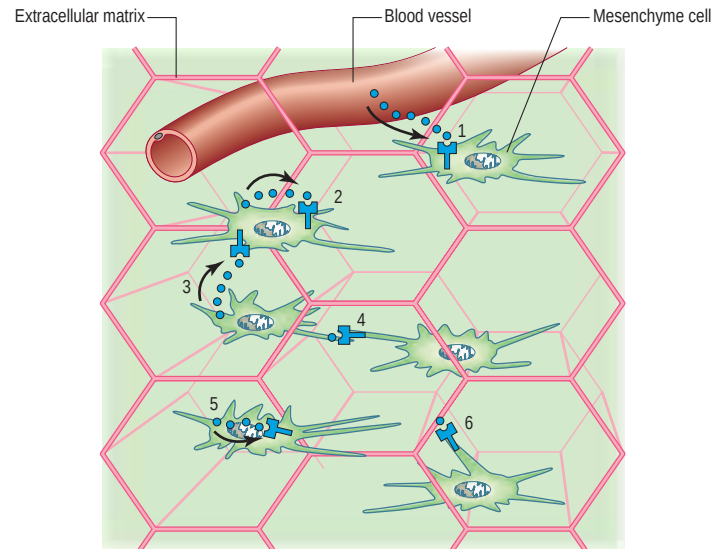
MORPHOGENESIS AND PATTERN FORMATION

Morphogenesis may be described as the assumption of form by the whole, or part, of a developing embryo. As a term, it is used to denote the movement of cell populations and the changing shape of an embryo, particularly during early development.

The most obvious examples of morphogenesis are the large migrations that occur during gastrulation; local examples include branching morphogenesis, which occurs in the developing lungs and kidneys, for example, and in most glandular organs. The development of branches from a tubular duct occurs over a period of time. An interaction between the proliferating epithelium of the duct and its surrounding mesenchyme and extracellular matrix results in a series of clefts that produce a characteristic branching pattern (Fig. 11.3). During tubular and acinar development, hyaluronidase secreted by the underlying mesenchymal cells breaks down the basal lamina produced by the epithelial cells; this increases epithelial mitoses locally and results in an expanding acinus. Cleft formation is initiated by the mesenchyme, which produces collagen III fibrils within putative clefts. (If the collagen is removed, no clefts develop, whereas if excess collagen is not removed, supernumerary clefts appear.) The collagen acts to protect the basal lamina from the effects of the hyaluronidase, which means that the overlying epithelia have a locally reduced rate of mitosis. The region of rapid mitoses

at the tip of the acinus is therefore split into two, and two branches develop from this point.

Pattern formation concerns the processes whereby the individual members of a mass of cells, initially apparently homogeneous, follow a number of different avenues of differentiation that are precisely related to each other in an orderly manner in space and time. The patterns embraced by the term apply not only to regions of regular geometric order, e.g. the crystalline lens, but also to asymmetric structures such as the limb. For such a process to occur, individual cells must be informed of their position within the embryo, and utilize that information for appropriate differentiation. Patterning of regions is seen in: the



- 1 Endocrine Delivery of growth factor by the blood stream from a distant biosynthetic site to the target mesenchyme cell.
- 2 Autocrine Synthesis of growth factor by the cell, its secretion, binding and activation of a surface receptor elsewhere on its own surface.
- 3 Paracrine Synthesis of growth factor by the cell, its secretion and diffusion to an adjacent cell (or group of cells) where it binds to and activates a cell surface receptor.
- 4 Juxtacrine Synthesis of growth factor by the cell. The growth factor remains on the cell surface and binds to and activates a receptor on an immediately adjacent cell.
- 5 Intracrine Synthesis of growth factor within the cytoplasm of the cell. The growth factor moves to the nucleus and binds and activates its own nuclear receptors.
- 6 Matricrine Synthesis and export of growth factor from the cell. The growth factor binds to the extracellular matrix where it remains active and subsequently binds to and activates a receptor for that growth factor on the same or a different cell.

Fig. 11.2 In addition to the mechanisms described in Figure 11.1, cells may also communicate by the reception, production and secretion of growth factors. A typical embryonic mesenchyme cell may receive and produce growth factors in this way.

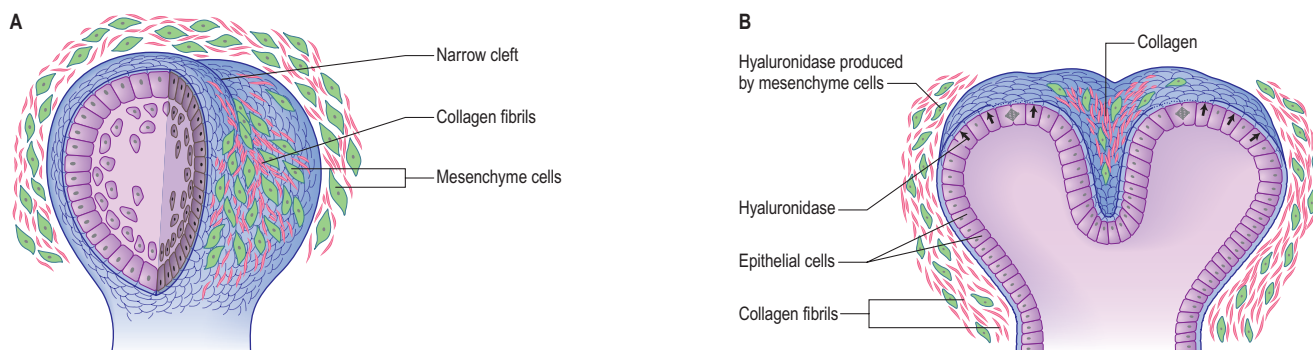


Fig. 11.3 Branching of a tubular duct may occur as a result of an interaction between the proliferating epithelium of the duct and its surrounding mesenchyme and extracellular matrix. **A**, Mesenchymal cells initiate cleft formation by producing collagen III fibrils locally within the development clefts and hyaluronidase over other parts of the epithelium. Collagen III prevents local degradation of the epithelial basal lamina by hyaluronidase and slows the rate of mitosis of the overlying epithelial cells. **B**, In regions where no collagen III is produced, hyaluronidase breaks down the epithelial basal lamina and locally increases epithelial mitoses, forming an expanded acinus (see arrows). (Adapted from Gilbert SF 1991 Developmental Biology. Sunderland, MA: Sinauer Associates.)

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progress zone and zone of polarizing activity within the limbs; the fates of the medial and lateral, and later the cranial and caudal, halves of the somites; and the neural crest mesenchyme within the pharyngeal arches. For details of patterning in vertebrate development, see [Tickle \(2003\)](#).

Genes in development

Two related themes have emerged from experimental studies of development: first, that the control of embryonic morphology has been highly conserved in evolution between vertebrates and invertebrates; and second, that this control involves families of genes coding for proteins that act as transcriptional regulators.

Homeobox genes are believed to be responsible, at least in part, for the evolutionary origin of the embryonic body plan ([Robert 2001](#)). Experimental study of transgenic animals in which the homeobox genes have been knocked out provide some evidence of their function; however, because developmental processes permit significant recovery from insult, some of the outcomes cannot be directly interpreted as demonstrating the effect of such gene loss.

Other gene families required for normal development include the T-box family, Helix-Loop-Helix transcription factors and *Sox* genes; the signalling factors, transforming growth factor beta (TGF- β), bone morphogenetic proteins (BMPs), fibroblast growth factor family (FGF), the

Wnt family and hedgehog signalling molecules. A range of cell receptor molecules are also required. For details on individual members within this range of factors, see [Carlson \(2014\)](#).

Experimental approaches to embryology

One of the most exciting techniques to provide information on cell movements and fates during development is the use of chimeric embryos. Small portions of an embryo are excised and replaced with similar portions of an embryo from a different species at the same stage, and the resulting development is then studied. This technique has been particularly effective using chick and quail embryos because the nucleolus is especially prominent in all quail cells, whereas it is not prominent in chick cells, which means that quail cells may be easily identified within a chick embryo after chimeric transplantation ([Le Douarin 1969](#)). The technique has also confirmed the reciprocity of tissue interaction between the embryonic species, a phenomenon that had previously been illustrated, for a limited period, in co-cultures of embryonic avian and mammalian tissues. Somite development and vertebral formation have been studied in mouse–chick chimeras ([Fontaine-Pérus 2000](#)). The production, *in vitro*, of human–animal chimeric cell lines is providing new ways of studying cellular pathways, as is the introduction of human artificial chromosome vectors into animal cells to study their interaction.

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